

4.19 Quiz: Probability (Algebra) — Markscheme

1. Events A and B are independent. It is given that $P(A) = 2P(B)$ and $P(A \cup B) = 0.72$.

Let $x = P(B)$.

(a) Show that $2x^2 - 3x + 0.72 = 0$. [3]

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (\text{M1})$$

$$\text{Since independent: } P(A \cap B) = P(A) \cdot P(B) = 2x \cdot x = 2x^2 \quad (\text{M1})$$

$$2x + x - 2x^2 = 0.72$$

$$2x^2 - 3x + 0.72 = 0 \quad (\text{AG})$$

Note: final line must be shown with algebraic working, not just stated. (A1)

(b) Hence find $P(B)$. [2]

$$x = \frac{3 \pm \sqrt{9 - 4(2)(0.72)}}{2(2)} = \frac{3 \pm \sqrt{3.24}}{4} = \frac{3 \pm 1.8}{4} \quad (\text{M1})$$

$$x = 1.2 \text{ (reject, } > 1) \text{ or } x = 0.3$$

$$P(B) = 0.3 \quad (\text{A1})$$

(c) Find $P(A \cap B)$. [1]

$$P(A \cap B) = P(A) \cdot P(B) = 0.6 \times 0.3 = 0.18 \quad (\text{A1})$$

2. In a school, 80 students were surveyed about activities A and B .

- $P(A) = 0.5$
- $P(B | A) = 0.3$
- $P(A' \cap B') = 0.2$

(a) Find $P(A \cap B)$. [2]

$$P(A \cap B) = P(B | A) \times P(A) \quad (\text{M1})$$

$$= 0.3 \times 0.5 = 0.15 \quad (\text{A1})$$

(b) Find $P(B)$. [2]

$$P(A \cup B) = 1 - P(A' \cap B') = 1 - 0.2 = 0.8 \quad (\text{M1})$$

$$P(B) = P(A \cup B) - P(A) + P(A \cap B) = 0.8 - 0.5 + 0.15 = 0.45 \quad (\text{A1})$$

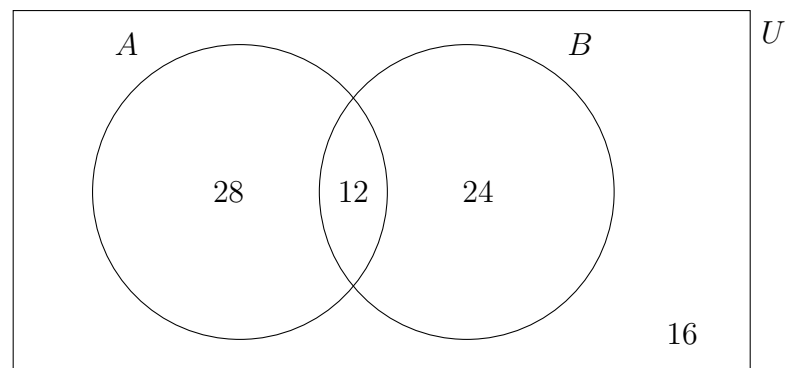
(c) Complete the Venn diagram below, showing the number of students in each region. [2]

$$n(A \text{ only}) = 80 \times (0.5 - 0.15) = 80 \times 0.35 = 28 \quad (\text{A1})$$

$$n(A \cap B) = 80 \times 0.15 = 12$$

$$n(B \text{ only}) = 80 \times (0.45 - 0.15) = 80 \times 0.30 = 24$$

$$n(\text{neither}) = 80 \times 0.2 = 16 \quad (\text{A1})$$



(d) Determine whether events A and B are independent. Justify your answer. [2]

$$P(A) \times P(B) = 0.5 \times 0.45 = 0.225 \quad (\text{M1})$$

$$P(A \cap B) = 0.15$$

$0.225 \neq 0.15$, so the events are **not independent**. (R1)

Total: 14 marks